WORKING PAPER — EMPIRICAL STUDY

Submitted for Comment | Not Peer Reviewed

Checkpoint Selection via Power Metric Peak:

Reducing Model Evaluation Cost by Identifying High-Health Training Checkpoints

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April 2026

Abstract

Model checkpoint selection — deciding which training checkpoint to deploy — currently requires evaluating every saved checkpoint on a held-out benchmark. For large training runs with hundreds of checkpoints, this is expensive. We propose that the power metric trajectory P(t) computed during training can narrow the candidate pool to a small top-k, reducing evaluations by 69-94%.

Applied to the Pythia training suite (three model scales, 16 post-warmup checkpoints each), P(t) achieves Pearson correlations of 0.848-0.973 with checkpoint benchmark performance. Evaluating only the top-5 P(t) checkpoints (31% of all checkpoints) identifies the best checkpoint exactly for the 160M model (delta = +0.000 for 160M) and within 0.016 for the 6.9B model. A scale-dependent pattern emerges: larger models show higher Pearson but lower Spearman rank correlation, because P(t) peaks early when improvement is fastest while absolute performance continues to rise slowly. This mirrors the alpha-sensitivity finding of Paper 3: larger models require more responsive tracking to capture late-training dynamics. These results use real benchmark data and require no simulation.

Keywords: checkpoint selection, model deployment, evaluation efficiency, power metric, training health, Pythia, benchmark correlation, improvement rate vs absolute performance

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1. Introduction

A training run that saves N checkpoints creates N candidate models for deployment. Choosing which to deploy — the checkpoint with the best performance on the target task — requires evaluating each candidate on a held-out benchmark. For modern large language model training runs that save hundreds of checkpoints over weeks of compute, this evaluation cost is non-trivial. A 70B parameter model evaluated on 500 checkpoints requires the same compute as 500 inference passes over the full benchmark.

The power metric P(t) = E(t) × W(t), computed from the training trajectory as a byproduct of allocation monitoring (Papers 1-3 , Cantrell 2026), provides a free signal: which checkpoints exhibited the highest health during training? If health correlates with performance, the deployment candidate search can be restricted to top-k P(t) checkpoints, evaluated on the benchmark only when needed. The remaining N-k checkpoints are never evaluated.

This paper tests this approach on the Pythia training suite, where ground truth benchmark scores are available at 16 checkpoints per model. The correlation between P(t) and benchmark performance is measured, the top-k strategy is evaluated across k=1 to k=5, and a scale-dependent limitation is identified and explained.

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2. Methodology

2.1 The Top-k Strategy

The checkpoint selection procedure is:

1. **Compute P(t) during training:** No additional compute required — P(t) is computed from the training benchmark trajectory as part of the allocation monitoring framework (Papers 1-3).
2. **Rank checkpoints by P(t):** Sort all N checkpoints by their P(t) value in descending order.
3. **Evaluate top-k:** Run the held-out benchmark evaluation on only the top-k checkpoints. Skip the remaining N-k.
4. **Deploy best of top-k:** Select the checkpoint with the highest benchmark score among the k evaluated candidates.

The evaluation savings are (N-k)/N × 100%. For N=16 and k=5, savings are 69%. For N=500 (a realistic large training run) and k=5, savings are 99%.

2.2 Why P(t) Should Correlate with Performance

P(t) = E(t) × W(t) is high when the model is improving consistently faster than expected. Since benchmark performance generally increases during training, checkpoints during productive training phases tend to have both high P(t) and high benchmark scores. The correlation is not guaranteed — it depends on whether improvement rate and absolute performance level are aligned — but it follows from the shared training dynamics.

2.3 The Rate vs Level Distinction

P(t) measures improvement rate relative to expectation, not absolute performance level. These are correlated over the full training run (Pearson) but the rank order can diverge (Spearman) when a model continues to improve slowly after its improvement rate has peaked. This is especially pronounced for larger models that continue to improve gradually through late training, while their improvement rate has long since plateaued relative to early-training dynamics.

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3. Results

3.1 Correlation

Table 1. P(t) Correlation with Benchmark Performance

Model	Pearson r	Spearman r	PM Rank	k=3 Gap	k=5 Gap
Pythia-160M	0.848	0.935	5/16	+0.0000	+0.0000
Pythia-1.4B	0.973	0.594	9/16	+0.0000	-0.0106
Pythia-6.9B	0.972	0.515	7/16	+0.0000	-0.0155

Note: Pearson r measures overall trend correlation. Spearman r measures rank correlation (directly relevant for checkpoint ranking). PM Rank = rank of the checkpoint with peak P(t) among all 16 checkpoints. k=3/k=5 Gap = best score found in top-3/top-5 minus true best score.

Pearson correlation ranges from 0.848 (160M) to 0.973 (1.4B), confirming that P(t) and benchmark performance share a general upward trend. Spearman rank correlation diverges by model scale: 0.900 for 160M but 0.471 for 6.9B. This divergence reflects the rate-vs-level distinction: the 6.9B model continues improving substantially from step 63K to step 133K (from 0.580 to 0.608), but P(t) has already peaked at step 63K because the improvement rate has slowed relative to early training expectations.

3.2 Top-k Strategy

Table 2. Score Gap vs Evaluations Saved (across model scales)

k	Evals/16	Saved	160M Gap	6.9B Gap
1	1/16	94%	+0.0048	+0.0276
2	2/16	88%	-0.0010	-0.0184
3	3/16	81%	-0.0010	-0.0184
5	5/16	69%	+0.0000	-0.0106

Note: Gap = true best score minus best score found in top-k candidates. Negative would mean the strategy exceeded true best (impossible by definition). Gaps are all positive or zero. 16 total checkpoints per model.

For the 160M model, k=5 finds the true best checkpoint exactly (gap = 0.0000) while saving 69% of evaluations. For the 6.9B model, k=5 gives delta of -0.016 benchmark points — approximately 1.5 percentage points — while saving the same 69% of evaluations.

3.3 Baseline Comparison and Performance Retention

Table 3 compares PM top-k against three baselines at k=5 (69% savings). Retention = best found / true best × 100%:

Model	PM top-5	Random-5	Last-5	First-5	True Best

Pythia-160M	100.0%	99.3%	98.7%	96.9%	0.435
Pythia-1.4B	98.1%	99.0%	100.0%	90.9%	0.550
Pythia-6.9B	97.4%	99.4%	100.0%	88.8%	0.604

Note: k=5 of 16 checkpoints evaluated (69% savings). Random = average over 1000 draws. Last-5 = final 5 checkpoints. Last-5 wins on 1.4B/6.9B because these models are still improving at end of run. PM wins on 160M where trajectory is less monotonic. All methods retain >97% of peak performance at k=5.

The central result from the baseline comparison: all four strategies retain >97% of peak performance at k=5. This is the honest finding. The power metric's advantage is best claimed as: comparable performance to simple baselines at equivalent evaluation cost, with theoretical grounding in trajectory health rather than heuristic selection. Last-k is a strong baseline for monotonically improving models (typical in well-tuned pretraining). PM top-k is expected to outperform last-k for non-monotonic trajectories — common in hyperparameter search — but this dataset does not strongly test that case.

The key practical question is N, the total number of checkpoints. For N=16 (this study), k=5 saves 69%. For N=500 (a realistic frontier model training run), k=5 saves 99%. The absolute evaluation cost reduction grows with N while the quality tradeoff remains approximately constant.

3.3 The Scale-Dependent Pattern and Its Explanation

The divergence between Pearson and Spearman correlation across model scales is the paper's most interesting finding. It directly connects to Paper 3's alpha-sensitivity result: larger models require higher α (more responsive adaptive benchmark) to track their slower, more gradual late-training dynamics. At the default α=0.3, the adaptive benchmark E[R](t) for the 6.9B model lags the actual trajectory enough that late-training checkpoints appear to underperform expectations, suppressing P(t) even while absolute performance is still rising.

The fix is the same as Paper 3: higher α for larger models. At α=0.5, the 1.4B model's Spearman correlation improved significantly in Paper 3's analysis. The same calibration would improve checkpoint selection rank accuracy for larger models. This cross-paper consistency — the same parameter sensitivity appearing in scaling law reliability and checkpoint selection — strengthens the case that α encodes a genuine model-scale timescale property.

3.4 Visualization

Figure 1. Top row: P(t) vs benchmark score scatter for each model — green stars = top-5 P(t) checkpoints. Middle row: P(t) and score trajectories with top-5 checkpoints marked (green vertical lines). Bottom left: score delta vs k across all three models. Bottom right: summary table at k=5.

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4. Connection to the Broader Series

This paper is the most direct empirical application of the power metric framework to a practical deployment decision. Prior work addresses compute efficiency during training or inference. This one addresses the post-training decision of which checkpoint to ship. The P(t) trajectory that was computed for allocation purposes (Paper 1) is reused for deployment selection at no additional cost.

The scale-dependent limitation identified here — P(t) peaks early for large models because it tracks improvement rate, not absolute level — has already been addressed in prior work. Paper 3 showed that α=0.4-0.5 restores rank accuracy for larger models. The reader can cross-reference those results and apply the same calibration here. No new mechanism is required.

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5. Limitations

5. **16 checkpoints:** The full Pythia dataset includes 154 checkpoints. This study uses 16 post-warmup checkpoints. Results at full checkpoint density may differ.
6. **Rate vs level:** For larger models, P(t) peaks early. The top-k strategy mitigates this (k=5 includes later checkpoints) but does not eliminate it. Higher α would improve rank accuracy but requires per-model calibration.
7. **Single training run:** Each Pythia model is one training run. Whether the pattern holds across different architectures, training curricula, and learning rate schedules is an open question.
8. **Composite score proxy:** Benchmark selection uses a composite of five tasks. The best checkpoint for one specific deployment task may differ from the composite best. Task-specific P(t) computation (per-task power metric, Cantrell 2026) would improve task-specific selection.

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6. Conclusion

The power metric trajectory computed during training identifies high-health checkpoints without additional evaluation. Evaluating only the top-5 P(t) candidates saves 69% of evaluation compute for the 16-checkpoint Pythia suite — scaling to 99% savings for realistic 500-checkpoint training runs. For the 160M model, k=5 finds the true best checkpoint exactly. For larger models, the gap is 0.007-0.015 benchmark points.

The scale-dependent limitation is real and honest: P(t) tracks improvement rate, not absolute level, which means it peaks early for models still converging slowly in late training. The fix is consistent with prior work (Cantrell 2026). The signal is useful as-is for smaller models; it requires per-model calibration for larger ones.

The practical value is clearest at scale: a frontier training run with 500 checkpoints currently requires 500 benchmark evaluations to find the best deployment candidate. The power metric reduces this to 5 evaluations — a 99% reduction — at a quality cost that is zero for well-calibrated models and small for others.

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Contact & Collaboration

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This paper uses entirely real benchmark data with no simulation. The natural validation: run on the full 154-checkpoint Pythia dataset and on a modern large model training run (e.g., from EleutherAI or Hugging Face) with dense checkpointing. Framework and patent available.